PCLDet: Prototypical Contrastive Learning for Fine-Grained Object Detection in Remote Sensing Images

Lihan Ouyang, Guangmiao Guo, Leyuan Fang, *Senior Member, IEEE*, Pedram Ghamisi, *Senior Member, IEEE*, and Jun Yue

***Abstract*— The capacity of satellites to supply high-resolution imaging has promoted the fine-grained object detection task in remote sensing images. However, this type of object detection is challenging due to low interclass feature differences in objects. To address this issue, we propose a prototypical contrastive learning-based detector (PCLDet) for fine-grained object detection in remote sensing images. The PCLDet first introduces the prototype to learn the fine-grained objects' features and then adopts contrastive learning to compare the target and the learned features, thus improving the differentiability of the fine-grained object. Specifically, we first introduce the prototype, which represents the feature centers of each class, and then construct a prototype bank to store the feature prototypes of each class. Then, we introduce contrastive learning to extract the discriminative features by maximizing the interclass distance and minimizing the intraclass distance. Furthermore, we propose the ProtoCL loss as a part of the model optimization, which enables more representative prototypes to be learned. Finally, to address the long-tail problem in the remote sensing fine-grained object detection dataset, we propose a new proposal sampler, the class-balanced sampler (CBS) that can sample each class equally. Extensive experiments demonstrate that our method can achieve the state-of-the-art performance on a commonly used aerial fine-grained object dataset (Fair1M) and aerial fine-grained ship dataset (OFSD) while maintaining high efficiency. The code will be available at https://github.com/G-Naughty/PCLDet.**

***Index Terms*— Class-balanced sampler (CBS), fine-grained object detection, prototypical contrastive learning, remote sensing images.**

Manuscript received 27 February 2023; revised 27 May 2023; accepted 18 June 2023. Date of publication 27 June 2023; date of current version 7 July 2023. This work was supported in part by the National Natural Science Foundation of China under Grant U22B2014 and in part by the Science and Technology Plan Project Fund of Hunan Province under Grant 2022RSC3064. *(Lihan Ouyang and Guangmiao Guo contributed equally to this work.) (Corresponding author: Leyuan Fang.)*

Lihan Ouyang is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China, and also with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710126, China (e-mail: oylh@hnu.edu.cn).

Guangmiao Guo is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China (e-mail: guoguangmiao@hnu.edu.cn).

Leyuan Fang is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China, also with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710126, China, and also with the Peng Cheng Laboratory, Shenzhen 518000, China (e-mail: fangleyuan@gmail.com).

Pedram Ghamisi is with the Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Helmholtz Institute Freiberg for Resource Technology, 09599 Freiberg, Germany, and also with the Institute of Advanced Research in Artificial Intelligence (IARAI), 1030 Vienna, Austria (e-mail: p.ghamisi@gmail.com).

Jun Yue is with the School of Automation, Central South University, Changsha 410083, China, and also with the Department of Geomatics Engineering, Changsha University of Science and Technology, Changsha 410114, China (e-mail: jyue@pku.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3290091

I. Introduction

OBJECT detection plays an important role in remote sensing image interpretation. It is a fundamental Earth observation task that aims at locating and classifying the object of interest [1]. Remote sensing object detection has been widely used for both military and civilian applications [2], [3], [4], [5], [6], [7], including geographic information system mapping [8], [9], urban monitoring [3], and traffic planning [7], [10].

In recent years, many convolutional neural network (CNN)-based methods [3], [11], [12], [13], [14], [15], [16], [17], [18], [19] have achieved promising performances for remote sensing object detection. Most of these methods [3], [11], [12] are motivated by corresponding methods for natural images. Object detection methods can be roughly summarized into one-stage [20], [21], [22], [23], [24] and two-stage [25], [26], [27], [28], [29] approaches. One-stage detectors directly perform bounding box regression and object classification, while two-stage detectors require two key steps: 1) generating proposals and 2) refining proposals and classifying them. The classical representatives of these two approaches are the YOLO series [20], [23], [24] and the regions with CNN feature (RCNN)-based methods [25], [26], [27], [28], [29]. To detect objects with uncertain orientation in remote sensing images, researchers have proposed oriented object detectors [1], [3], [12], [13], [14], [15], [16], [17], [18], [19], [30] based on the traditional one- and two-stage object detectors. The one-stage-based oriented object detectors [1], [15], [18], [19] are characterized by fast inference due to the simplicity of the one-stage detector framework. The typically oriented object detectors based on one-stage detectors [15], [18], [19] are single-shot alignment network (S^2A-Net) [18], R3Det [15], SCRDet [19], and others like them. The two-stage-based oriented object detector [13], [14], [16], [17] is more concerned on detection accuracy. This type of oriented detectors (e.g., RoI Transformer [14], ReDet [17], and Oriented RCNN [13]) evolved from two-stage detectors. Although many methods

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

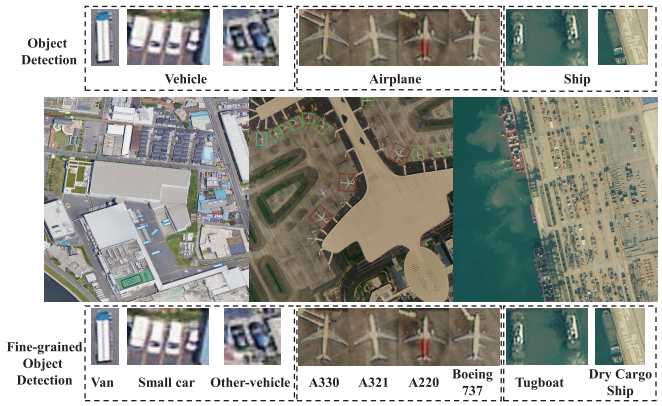


Fig. 1. Comparison between ordinary remote sensing target detection and fine-grained remote sensing target detection.

have been proposed for object detection in remote sensing images, fine-grained object detection in remote sensing images still remains a challenge.

Because the resolution of remote sensing images is higher, fine-grained object detection in remote sensing has become possible. Fine-grained object detection in remote sensing has a variety of applications, such as national defense security, battlefield environment awareness, and facilities management [31]. Compared with general object detection, fine-grained object detection methods classify objects with greater detail [32]. As shown in Fig. 1, fine-grained object detection not only classifies the course category (e.g., vehicle, airplane, and ship) but also recognizes that the object belongs to a refined subcategory (e.g., van, A330, Boeing 737, and tugboat). The comparisons in Fig. 1 show a slight interclass difference between fine-grained subcategories, which limits the performance of the abovementioned detectors for fine-grained remote sensing object detection.

Very recently, some new remote sensing fine-grained object recognition methods have been developed. Works in [33], [34], and [35] utilize attention mechanisms to focus on discriminative parts in low discriminative objects to help recognition. For instance, Sumbul et al. [34] introduced the attention-driven deep feature representations of multisource to classify the fine-grained targets. Although the above methods introduce new mechanisms for fine-grained object recognition and obtain promising performance, the fine-grained object features extracted are not highly discriminated and can hardly be accurately distinguished.

To address the above issue, we propose a novel method, prototypical contrastive learning-based detector (PCLDet), for fine-grained object detection in remote sensing images. First, we introduce prototypes that represent the feature centers of each class, then build a prototype bank that stores the feature prototypes of each class. Second, we introduce contrastive learning to extract discriminative features by widening the interclass distance and narrowing the intraclass distance. Third, we design a ProtoCL loss to optimize the model and assist in learning more representative prototypes. Finally, to tackle the long-tail problem in the remote sensing fine-grained object detection dataset, we propose a new sampler, the class-balanced sampler (CBS) that can sample each class equally.

Our key contributions are given as follows.

1) The prototype bank and the ProtoCL loss are proposed to extract discriminative features by maximizing the interclass distance and minimizing the intraclass distance. The discriminative features obtained enable efficient fine-grained object detection.
2) We propose the CBS to alleviate the long-tail problem in remote sensing fine-grained datasets. The purpose of the CBS is to sample an equal number of positive samples from each category in the region proposal network (RPN).
3) We conduct comprehensive experiments on the remote sensing fine-grained detection datasets Fair1M [32] and OFSD [36] to evaluate the performance of our method. The results demonstrate that the PCLDet outperforms other state-of-the-art (SOTA) methods with respect to fine-grained detection in remote sensing images.

The rest of this article is organized as follows. We briefly introduce the related work in Section II. Section III describes our proposed PCLDet, including the prototype bank, the ProtoCL loss, and the CBS. Next, the comprehensive experiments and ablation experiments are conducted in Section IV. Finally, Section V concludes this article.

II. Related Work

In this section, we briefly review works in three related domains: remote sensing object detection, fine-grained object recognition, and contrastive learning.

A. Remote Sensing Object Detection

Object detection in remote sensing images has attracted increasing attention in recent years [13]. Recent mainstream object detection methods can be roughly summarized into one-stage [20], [21], [22], [23], [24] and two-stage [25], [26], [27], [28], [29] approaches. The classical one-stage detectors are the YOLO series [20], [23], [24], SSD [21], RetinaNet [22], and others like them. One-stage detectors directly predict the location and class of the object through a dense network. Due to the simplicity of the one-stage detector [20], [21], [22], [23], [24] framework, its inference is fast, but detection accuracy is not high. Two-stage detectors [25], [26], [27], [28], [29] contain an additional candidate region generation stage and are more concerned with detection accuracy. RCNN [29] is the representative two-stage detector. The classical two-stage detectors, Faster RCNN [25], Mask RCNN [28], and Cascade RCNN [26], have all evolved from RCNN [29].

Some works [3], [11] apply one- and two-stage detectors to remote sensing fields. Due to the indeterminate orientation of the object in remote sensing images, researchers have proposed oriented detectors [13], [14], [15], [16], [17], [18], [19] based on one- and two-stage detection methods. Recently, works in [15], [18], and [19] have proposed an oriented object detector based on the one-stage detection method. Yang et al. [15] proposed a refined one-stage oriented object detector, which involves two key improvements, including feature refinement

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

and progressive regression, to address the issue of feature misalignment. Han et al. [18] proposed an S^2A-Net for oriented object detection that aims at improving location accuracy via deep feature alignment. Although one-stage detectors [15], [18], [19] often have faster speeds, their accuracy is lower than two-stage detectors. Several works [13], [14], [16], [17] have proposed oriented object detectors based on two-stage detection methods. For example, Ma et al. [37] presented rotated RPNs (RRPNs) to achieve oriented object detection by setting rotatable anchors in the RPN phase. Ding et al. [14] built the RoI Learner to learn rotated regions of interest (RoIs) from horizontal RoIs produced by RPN. Xie et al. [13] proposed Oriented RCNN to obtain oriented representation by learning midpoint offsets.

B. Fine-Grained Object Recognition

In general, works for fine-grained object recognition can be divided into three categories: localization-based, attention-based, and feature-encoding-based [38], [39], [40]. Localization-based methods [41], [42], [43] classify objects by learning information about their distinguishing locations (e.g., the beak of a bird). Part-based R-CNN [43] employs the detection network RCNN to locate the head or body of birds and then uses the located regions for classification. Attention-based methods [44], [45] utilize powerful properties of attention mechanisms to learn discriminative regions. Behera et al. [46] proposed CAP, which introduces context-dependent attention features guided by regional features to obtain good performance in fine-grained image recognition tasks. Feature-encoding-based methods [38], [39], [40] learn more discriminative features through high-dimensional feature interactions. For example, Yu et al. [40] proposed a hierarchical bilinear pooling framework to learn interlayer part feature interactions from the intermediate convolutional layers, achieving mutual reinforcement of interlayer feature interactions and differentiated part feature learning.

In the remote sensing field, some attention-based fine-grained object recognition methods have made great progress [33], [34], [35]. Work [34] classifies scenes by locating multiscale discriminative regions through attention networks. The RNN-based attention mechanism [35] is introduced to recognize fine-grained ships.

C. Contrastive Learning

Contrastive learning has been shown to be a superior method in feature representation [47] and has also achieved outstanding performance in unsupervised learning [48], few-shot learning [43], and face recognition [49], [50]. For example, few-shot object detection via contrastive proposals encoding (FSCE) [51] constructs a supervised contrastive head to extract more discriminative features in few-shot learning. Supervised contrastive learning [52] extends the method of batch contrast to a supervised setting. Due to the unique idea of contrastive learning, researchers have also applied it to fine-grained image recognition task [31], [53]. For instance, Li et al. [53] proposed BSNet, which uses two similarity measures to enable the model to learn more discriminative and less similarity-biased features.

III. Methodology

In this section, we first introduce the framework of the PCLDet. Second, we propose the CBS module to alleviate the long-tail problem in fine-grained datasets. Third, we construct a prototype bank to store and update the feature prototypes with ProtoCL loss. Finally, we introduce the loss functions utilized in the PCLDet.

A. Overall Architecture

The overall framework of our proposed PCLDet is shown in Fig. 2. In this section, the baseline we choose is the ReDet [17], which consists of a backbone, ReFPN, RPN, and detection head. ReResNet50 is adopted as the backbone to extract the features. As shown in Fig. 1, the slight interclass difference between fine-grained subcategories limits the performance of fine-grained object detection in remote sensing images. Very recently, contrastive learning has been shown to perform well in feature representation [47]. The core idea of contrastive learning is expanding the interclass distance and narrowing the intraclass distance. Motivated by it, we propose a PCL module that contains a prototype bank and a ProtoCL loss to extract discriminative features. In the PCL module, we first introduce the prototypes to represent the center feature of each class and construct a prototype bank to store them. Then, we achieve discriminative feature extraction by decreasing the distance between the sample and the corresponding prototype while increasing the distance between the sample and the other prototypes. Finally, the ProtoCL loss is used to update the prototypes. In addition, we propose the CBS to alleviate the long-tail issue in the remote sensing fine-grained object detection dataset, as shown in Fig. 3.

Both the CBS and the PCL modules proposed by the PCLDet are plug-in modules that can be used in each two-stage detection method.

B. Prototypical Contrastive Learning Module

As shown in Fig. 1, the feature similarity between fine-grained objects is high. Contrastive learning extracts the discriminative features by maximizing the interclass distance and minimizing the intraclass distance [51]. Motivated by it, we introduce contrastive learning into remote sensing fine-grained object detection. The key point of contrastive learning is selecting the positive and negative samples. In this article, the prototype of the class to which the sample belongs is defined as positive and the prototype of the other categories is defined as negative.

Specifically, a prototype bank $[\mathbf{w_0}, \mathbf{w_1}, \ldots, \mathbf{w_n}]$ is built in the PCL module (n represents the number of classes). At the beginning of the training, we initialize the prototype features with the normal distribution. As shown in Fig. 2, for each input feature x, the PCL module encourages x to be closer to the corresponding prototype feature while staying away from other categories of prototypes. We adopt cosine similarity to measure the distance between the input feature and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

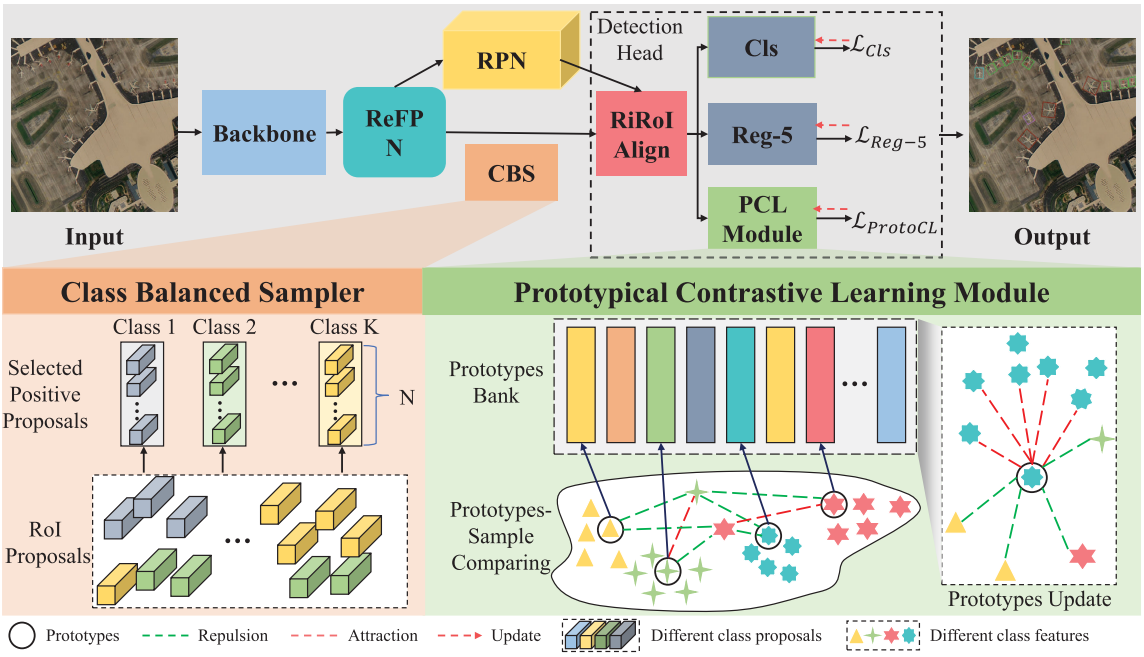


Fig. 2. Framework of the proposed PCLDet. For input remote sensing images, the PCLDet first applies the ReResNet50 backbone and ReFPN in order to extract features. Then, the PCLDet utilizes the CBS in RPN to generate proposals. After that, the PCL module increases fine-grained target interclass distances by contrasting prototype and sample features. Finally, the PCLDet updates the prototypes in the prototypes bank with the ProtoCL loss.

prototypes, which is defined as follows:

$$d_i = \frac{w_i x}{|w_i||x|}. \tag{1}$$

Then, we adopt the ProtoCL loss to constrain the interclass feature distance and optimize the feature representation of samples. The detail of ProtoCL loss is defined as follows:

$$\begin{aligned} L_{\text{Pro}} &= \frac{-1}{|D^+|}\sum_{k\in K}\log\frac{\left(\sum_{y_i=k} e^{\frac{d_i}{\tau}}\right)}{\left(\sum_{y_i\neq k} e^{\frac{d_i}{\tau}}\right)} \\ &= \underbrace{\frac{-1}{|D^+|}\sum_{k\in K}\log\left(\sum_{y_i=k} e^{\frac{d_i}{\tau}}\right)}_{\text{sample-prototype align}} + \underbrace{\frac{1}{|D^+|}\sum_{k\in K}\log\left(\sum_{y_i\neq k} e^{\frac{d_i}{\tau}}\right)}_{\text{sample-prototype away}} \end{aligned} \tag{2}$$

where $|D^+|$ is the number of categories appearing in the batch, K is the number of all categories in the dataset, k is the kth category in K categories, and y_i is the label of ith proposals. τ is a scalar temperature parameter, which is set to 1/32. We assume that the first term of the L_{Pro} is "sample-prototype align" and the second term of the L_{Pro} is "sample-prototype away." "Sample-prototype align" is used to pull the sample closer to the corresponding prototype, while sample-prototype away is used to push the sample away from other prototypes.

In the object detection task, the number of negative proposals is much higher than the number of positive proposals. The L_{Pro} term makes the model more focus on pushing away samples and other prototypes during the update while ignoring the alignment of samples and corresponding prototypes.

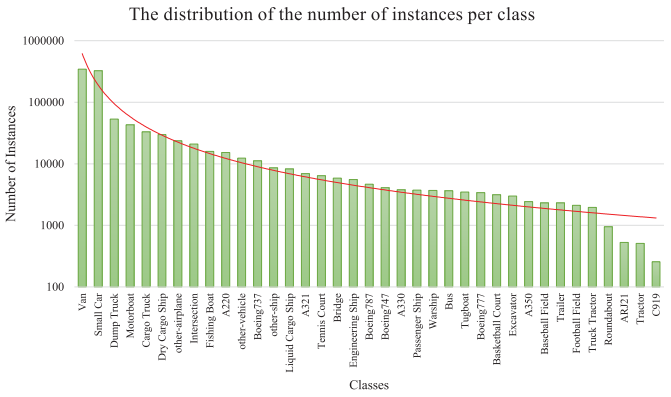


Fig. 3. Distribution of remote sensing fine-grained detection dataset Fari1M instances.

Therefore, we adjust L_{ProtoCL} as follows:

$$\begin{aligned} L_{\text{ProtoCL}} = \frac{1}{|D^+|}\sum_{k\in K}\log\left(1+\sum_{y_i=k}\exp\left(-d_{(i,k)}/\tau\right)\right) \\ +\frac{1}{K}\sum_{k\in K}\log\left(1+\sum_{y_i\neq k}\exp\left(d_{(i,k)}/\tau\right)\right). \end{aligned} \tag{3}$$

In the training phase, we optimize the model and learn a more representative prototype by L_{ProtoCL}. For each prototype w_i, in this article, we utilize stochastic gradient descent (SGD) to update it. The updated formula of prototypes is given as follows:

$$w_i^{t+1} = w_i^t - \alpha\frac{\partial}{\partial w_i^t}L_{\text{ProtoCL}} \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

where α represents the learning rate, which is determined in Section IV.

C. Class-Balanced Sampler

As shown in Fig. 3, the long-tail issue is obvious in fine-grained object detection for remote sensing images. It leads to the underfitting of the tail class during the training phase.

To alleviate the long-tail problem during the training phase, we propose the CBS, a simple but effective sampler method. In the RPN, the original sampler randomly samples proposals that are determined to be positive samples. The motivation of the CBS is to sample an equal number of positive samples from each category. Specifically, we define the proposals R_i whose intersection over union (IoU) with ground truth is higher than φ as positive samples R_{pos}, and others are defined as negative samples R_{neg}, as shown in the following:

$$\begin{cases} R_{\text{pos}} = R_i, & \text{if IoU}(R_i, \text{GT}) >= \varphi \\ R_{\text{neg}} = R_i, & \text{otherwise} \end{cases} \tag{5}$$

$$\text{IoU} = \frac{\text{Area}(R_i) \bigcap \text{Area(GT)}}{\text{Area}(R_i) \bigcup \text{Area(GT)}} \tag{6}$$

where Area() represents the area calculation (e.g., Area(R_i) represents the area of R_i) and GT represents the ground truth.

Suppose that we need to sample P positive samples from C categories in a batch. Unlike the original sampler that samples P samples randomly, the CBS sets a sampling index for each class based on the number of positive samples required and the number of classes in the batch, as shown in the following:

$$N = \frac{P}{C}. \tag{7}$$

We first sample as many N samples as possible for each class of targets. Then, if the number of positive proposals R^k_{pos} obtained by sampling is M and is less than N, random sampling is performed from the proposals R^k_{neg} that have not been sampled until the sampled number reaches N

$$\begin{cases} R^k_{\text{pos}} = R^j_i, & \text{if } j = k \\ R^k_{\text{neg}} = R^j_i, & \text{if } j \neq k. \end{cases} \tag{8}$$

As mentioned above, the CBS ensures that rare samples can be sampled in a balanced way when they appear, ensuring that rare categories are learned effectively in prototypical contrastive learning.

D. Loss Function

In this article, we use a multitask function to balance the classification and localization task loss. The total loss of the PCLDet is defined as follows:

$$L_{\text{total}} = L_{\text{rpn}} + L_{\text{ProtoCL}} + L_{\text{Reg-5}} + L_{\text{Cls}} \tag{9}$$

where L_{rpn}, L_{ProtoCL}, $L_{\text{Reg-5}}$, and L_{Cls} denote the RPN loss, ProtoCL loss defined in Section III-B, regression loss, and classification loss, respectively. In our proposed PCLDet, L_{rpn} is the same as defined in ReDet [17]. The regression parameters in $L_{\text{Reg-5}}$ are defined as

$$\begin{aligned} t_x &= \frac{1}{w}\left(\left(x' - x\right)\cos(\theta) + \left(y' - y\right)\sin(\theta)\right) \\ t_y &= \frac{1}{h}\left(\left(y' - y\right)\cos(\theta) - \left(x' - x\right)\sin(\theta)\right) \\ t_w &= \log\frac{w'}{w}, t_h = \log\frac{h'}{h} \\ t_\theta &= \frac{1}{2\pi}\left(\left(\theta' - \theta\right) \bmod 2\pi\right) \end{aligned} \tag{10}$$

where (x, y, w, h, θ) is the center point coordinates, width, height, and angle of a rotated region of interest (RRoI) and $(x', y', w', h', \theta')$ is the ground-truth parameters. The mod is used to ensure $t_\theta \in [0, 2\pi)$. Meanwhile, we employ the smooth L1 loss and cross-entropy loss as $L_{\text{Reg-5}}$ and L_{Cls}, and the detail of these two losses is given as follows:

$$L_{\text{Cls}} = -\frac{1}{N}\sum_i \sum_{c=1}^{K} y_{ic} \log p_{ic} \tag{11}$$

$$L_{\text{Reg-5}} = \begin{cases} 0.5t^2, & \text{if } |x| < 1 \\ |t| - 0.5, & \text{otherwise} \end{cases} \tag{12}$$

where N is the number of samples, K is the number of classes in the dataset, y_{ic} is a symbol function (if the ground truth of sample i is c, y_{ic} is 1; otherwise, it is 0), p_{ic} is the predicted probability of sample i belonging to the class c, and t represents the regression parameters defined above.

TABLE I
ABLATION STUDY ON OFSD

Method	PCL module	CBS	mAP(%)
Baseline			79.03
Baseline+PCL module	✔		80.43
Baseline+CBS		✔	79.38
PCLDet(Ours)	✔	✔	**81.2**

The bolded ones indicate the best results.

IV. EXPERIMENTS

A. Datasets

We used two datasets in our experiments: Fair1M [32] and OFSD [36].

1) Fair1M: This remote sensing image dataset for fine-grained object detection contains 42 796 images with sizes ranging from 600 to 10 000. It has 1.02 million instances within 37 categories: Passenger Ship, Motorboat, Fishing Boat, Tugboat, Engineering Ship, Liquid Cargo Ship, Dry Cargo Ship, Warship, Other ship, Small Car, Bus, Cargo Truck, Dump Truck, Van, Trailer, Tractor, Excavator, Truck Tractor, Other vehicle, Boeing 737, Boeing 747, Boeing 777, Boeing 787, ARJ21, C919, A220, A321, A330, A350, Other airplane, Baseball Field, Basketball Court, Football Field, Tennis Court, Roundabout, Intersection, and Bridge.

We use the training set for training and the test-benchmark set for testing. Horizontal flips, vertical flips, and diagonal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II
ABLATION STUDY ON FAIR1M

model name	PCL	CBS	Boeing 737	Boeing 747	Boeing 777	Boeing 787	C919	A220	A321	A330	A350	ARJ21	Passenger Ship
Baseline			36.51	87.03	22.45	50.68	10.82	47.92	71.7	72.45	77.57	38.8	**18.38**
Baseline +PCL	✔		**45.48**	86.83	25.34	47.74	21.47	51.16	69.42	73.64	75.47	**41.75**	15.56
Baseline +CBS		✔	44.44	**87.81**	24.64	49.97	**24.86**	49.64	69.84	73.91	74.61	41.6	17.27
PCLDet	✔	✔	43.51	87.37	**28.73**	**55.37**	20.04	**51.65**	**73.06**	**74.52**	**79.86**	31.08	18.21

model name	Motorboat	Fishing Boat	Tugboat	Engineering Ship	Liquid Cargo Ship	Dry Cargo Ship	Warship	Small Car	Bus	Cargo Truck	Dump Truck	Van	Trailer
Baseline	61.71	11.77	**35.87**	13.35	23.83	41.56	34.36	**61.82**	19.96	41.74	47.33	56.3	13.6
Baseline +PCL	59.22	10.71	33.64	16.59	**27.36**	41.57	32.28	58.85	32.38	39.74	45.48	51.83	**15.09**
Baseline +CBS	60.18	11.42	32.63	**18.09**	26	41.17	33.56	58.59	**33.14**	40.92	45.9	52.18	14.64
PCLDet	**62.88**	**12.05**	35.17	13.52	23.79	**42.46**	**37.45**	61.26	29.13	**42.81**	**47.99**	**56.93**	14.57

model name	Tractor	Excavator	Truck Tractor	Basketball Court	Tennis Court	Football Field	Baseball Field	Intersection	Roundabout	Bridge	mAP
Baseline	1.98	12.2	0.79	54.67	79.35	70.09	90.57	61.7	20.47	38.82	42
Baseline +PCL	4.83	11.76	**1.39**	**55.12**	**86.31**	66.05	**90.96**	61.06	18.74	**44.84**	42.93
Baseline +CBS	2.84	8.81	0.81	54.98	83.6	68.87	90.36	61.58	**22.7**	43.81	43.1
PCLDet	**5.19**	**13.08**	1.36	55.01	78.69	**69.55**	90.69	**62.35**	20.55	39.3	**43.5**

The bolded ones indicate the best results.

TABLE III
COMPARISON WITH SOTA METHODS ON FAIR1M

Course class	Fine-grained class	RetinaNet-O [22]	S^2A-Net [18]	Faster R-CNN-O [54]	Gliding Vertex [16]	ROI Trans [14]	ORCNN [13]	ReDet [17]	RB-FPN [55]	PCLDet (RoITrans)	PCLDet (ORCNN)	PCLDet (ReDet)
Airplane	Boeing737	33.01	33.32	37.11	36.32	37.30	37.08	36.51	39.16	40.16	36.36	**43.51**
	Boeing747	76.71	79.07	84.23	86.05	85.16	85.96	87.03	86.7	84.72	**87.44**	87.37
	Boeing777	11.52	18.08	15.64	11.96	16.23	13.38	22.45	16.17	20.31	16.33	**28.73**
	Boeing787	32.95	34.84	46.95	44.32	46.74	48.12	50.68	49.16	50.56	54.84	**55.37**
	C919	1.31	2.63	9.88	16.98	11.73	14.59	10.82	13.6	**20.36**	19.35	20.04
	A220	42.88	43.31	48.15	49.72	44.84	45.90	47.92	49.18	48.92	45.55	**51.65**
	A321	62.93	64.50	66.46	66.99	71.46	69.07	71.70	66.91	**73.15**	69.84	73.06
	A330	42.96	61.27	68.86	71.19	69.31	70.59	72.45	71.43	72.06	73.85	**74.52**
	A350	63.98	64.45	69.33	70.22	68.97	73.99	77.57	74.34	73.76	75.53	**79.86**
	ARJ21	2.72	3.22	25.23	28.43	33.92	28.29	**38.80**	29	33.77	34.48	31.08
	Passenger Ship	5.30	7.68	9.08	12.57	15.86	16.84	18.38	**18.38**	16.07	16.73	18.21
Ship	Motorboat	24.04	53.82	49.44	54.23	58.77	60.48	61.71	**68.41**	54.10	59.55	62.88
	Fishing Boat	3.03	9.00	4.55	7.12	8.63	10.17	11.77	10.55	8.27	9.28	**12.05**
	Tugboat	27.09	34.71	31.85	**39.02**	33.21	35.60	35.87	38.27	33.55	35.80	35.17
	Engineering Ship	7.38	7.97	9.31	12.34	12.84	12.04	13.35	11.83	13.01	12.69	**13.52**
	Liquid Cargo Ship	6.39	20.02	9.79	18.47	20.98	21.81	23.83	**25.12**	21.12	22.51	23.79
	Dry Cargo Ship	21.97	38.68	25.93	35.68	40.57	41.47	41.56	38.41	41.09	42.06	**42.46**
	Warship	4.60	25.55	10.42	16.77	27.58	28.99	34.36	34.72	31.11	30.58	**37.45**
	Small Car	38.86	62.73	54.38	54.78	59.81	58.85	61.82	**70.75**	56.15	58.44	61.26
	Bus	4.95	12.82	18.76	29.13	29.20	24.40	19.96	**36.14**	31.72	31.41	29.13
	Cargo Truck	19.03	34.61	35.85	39.39	41.97	42.31	41.74	**44.94**	42.05	43.49	42.81
	Dump Truck	13.00	34.85	41.29	43.05	46.99	46.83	47.33	**50.2**	45.73	47.03	47.99
Vehicle	Van	29.13	56.95	48.91	49.14	57.02	54.91	56.30	**70.77**	50.36	53.42	56.93
	Trailer	0.03	3.65	8.25	12.79	14.92	15.35	13.60	**16.75**	13.45	15.94	14.57
	Tractor	0.03	1.63	1.88	1.91	2.37	5.66	1.98	1.68	4.58	**5.70**	5.19
	Excavator	0.17	4.27	7.17	11.74	10.53	14.59	12.20	17.24	12.50	**18.33**	13.08
	Truck Tractor	0.00	0.56	0.40	0.63	0.49	0.57	0.79	0.49	**1.37**	0.95	1.36
	Basketball Court	27.52	41.98	48.93	46.10	49.80	49.17	54.67	54.59	50.64	52.95	**55.01**
Court	Tennis Court	79.65	78.95	78.58	80.11	79.96	81.09	79.35	80.49	78.49	**83.37**	78.69
	Football Field	55.09	58.87	53.55	52.78	63.65	63.42	**70.09**	65.69	62.46	64.86	69.55
	Baseball Field	86.32	87.65	87.80	87.01	88.54	89.91	90.57	88.92	89.16	88.52	**90.69**
	Intersection	52.81	50.63	59.15	58.23	57.90	60.12	61.70	56.64	56.99	60.18	**62.35**
Road	Roundabout	21.66	16.58	22.44	20.66	20.97	15.96	20.47	20.37	18.52	**23.28**	20.55
	Bridge	9.41	18.74	13.48	22.57	24.42	31.34	38.82	32.19	36.94	32.64	**39.30**
	mAP	26.72	34.34	35.38	37.89	39.78	40.26	42.00	42.62	40.80	41.86	**43.50**

ORCNN represents Oriented RCNN, RoI Trans represents RoI Transformer. The bolded ones indicate the best results.

flips are used as data arguments in the training phase. For a fair comparison, we adopt the same data arguments in all methods. The detection accuracy of the methods in this article is obtained by submitting results to the Fair1M benchmark evaluation server.

2) OFSD: OFSD is a high-resolution fine-grained ship detection dataset annotated with a rotated bounding box that contains 9451 images and 41 937 instances of 133 categories. We use the training set and validation set for training, and the testing set for testing. All images are resized to 1024 × 1024 in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON WITH SOTA METHODS ON OFSD

Method	RetiaNet-O [22]	S^2A-Net [18]	Faster-RCNN-O [54]	Gliding Vertex [16]	ORCNN [13]	RoI Trans [14]	ReDet [17]	PCLDet (RoI Trans)	PCLDet (ORCNN)	PCLDet (ReDet)
Input Size	1024	1024	1024	1024	1024	1024	1024	1024	1024	1024
Backbone	ResNet50	ResNet50	ResNet50	ResNet50	ResNet50	ResNet50	ReResNet50	ResNet50	ResNet50	ReResNet50
Framework	One	One	Two	Two	Two	Two	Two	Two	Two	Two
mAP(%)	21.16	47.01	49.45	64.06	73.53	74.43	79.03	76.28	77.48	**81.2**

ORCNN represents Oriented RCNN, RoI Trans represents RoI Transformer. The bolded ones indicate the best results.



Fig. 4. Some detection results of the proposed PCLDet on Fair1M.

the training and testing phases. Horizontal flips, vertical flips, and diagonal flips are used as data arguments in the training phase.

B. Implementation Details

Our experimental environment is the Pytorch in the Ubuntu 20.01 system, and all the experiments are performed on an NVIDIA RTX 3090 GPU with 24-GB memory. We adopt ResNet50 and ReResNet50 pretrained on ImageNet as a backbone network, the SGD algorithm to optimize the parameters of the model, and Kaiming normal to initialize layers. The initial learning rate, weight decay, and momentum of SGD are 0.005, 0.0001, and 0.9, respectively. We set the total epoch of all the experiments as 12 and reduce the learning rate at the 8th and 11th epochs. The batch size of the experiment is set as 2. The nonmaximum suppression (NMS) threshold is set as 0.1 and the IoU is set as 0.5 in all the experiments. We use mean average precision (mAP), which is commonly used in target detection tasks, as the evaluation metric.

C. Ablation Studies

To validate the effectiveness of each of our modules, we set up the following ablation experiments on the Fair1m and OFSD datasets. The experimental results demonstrate the effectiveness of the PCL module and the CBS.

1) Baseline Setup: Our baseline is ReDet [17], a two-stage oriented object detection method. It uses ReResNet50 as the backbone, cross-entropy loss as classification loss, and smooth L1 loss as regression loss. For a fair comparison, all hyperparameters are set the same in all experiments. As shown in Tables I and II, the baseline reaches 79.03 points and 42 points in mAP on OFSD and Fair1M, respectively.

2) Effectiveness of the PCL Module: The proposed PCL module in the PCLDet is designed to extract more discriminative features. To verify the effectiveness of the PCL module, we design a set of ablation experiments: baseline, baseline + PCL module, baseline + CBS, and the PCLDet.

As shown in Tables I and II, the baseline + PCL module surpasses the baseline by a big margin. Compared with the baseline, the baseline + PCL module improves mAP by 1.4 points and 0.93 points on OFSD and Fair1M, respectively, without consuming inference speed. Moreover, the PCLDet gains improvement of 1.82 points and 0.4 points on OFSD and Fair1M compared with baseline + CBS, as shown in Tables I and II, respectively. In addition, the baseline + PCL module improves the performance of the subcategories of airplanes

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

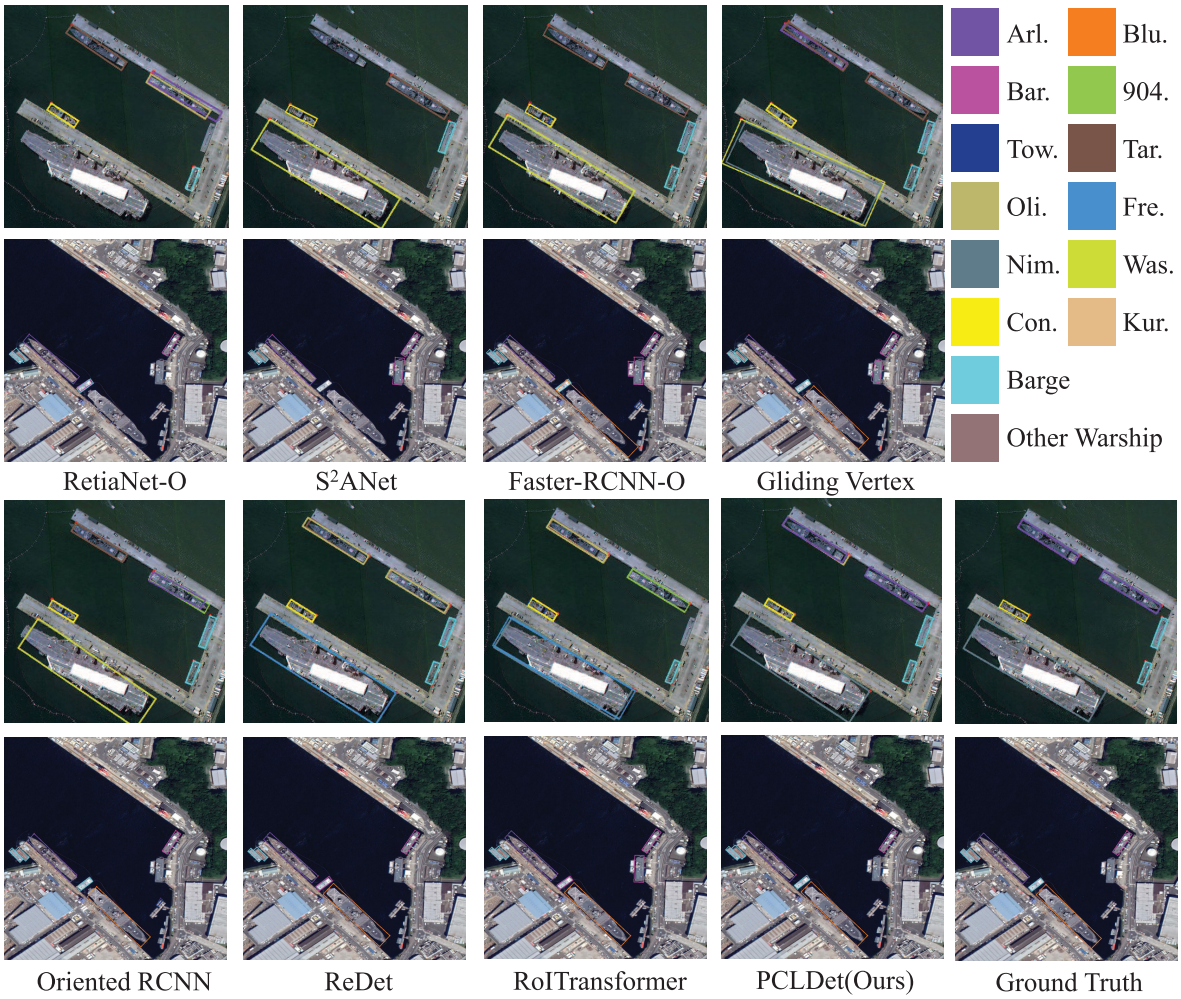


Fig. 5. Visualization of compared SOTA methods results on OFSD. The category abbreviations are given as follows: Arleigh Burke-class Destroyer (Arl.), Blue Ridge class command ship (Blu.), Barracks Ship (Bar.), Towing vessel (Tow.), Oliver Hazard Perry-class frigate (Oli.), Nimitz Aircraft Carrier (Nim.), Container Ship (Con.), Tarawa-class amphibious assault ship (Tar.), Freedom-class littoral combat ship (Fre.), Wasp-class amphibious assault ship (Was.), Kurobe-class training support ship (Kur.), and 904-general stores issue ship (904.)

and vehicles. These results illustrate that the PCL module contributes to the network for extracting the discriminative features.

3) Effectiveness of CBS: The proposed CBS in the PCLDet is designed to alleviate the long-tail issue. To verify the effectiveness of the CBS, we design a set of ablation experiments: baseline, baseline + PCL module, baseline + CBS, and the PCLDet. In the baseline and baseline + PCL module, we use the random sampler to replace CBS.

As shown in Tables I and II, the mAP is greatly improved when applying the CBS. We observed that the quantification results of baseline + CBS gain an improvement of 0.35 points and 1.1 points in the OFSD and Fair1M, compared to baseline. Moreover, the PCLDet achieves 81.2 points and 43.5 points in terms of mAP on OFSD and Fair1M, respectively, which is higher than the baseline + PCL module. On Fair1M, the baseline + CBS can obtain 24.86 points (14.04 points↑), 22.7 points (2.23 points↑), 14.64 points (1.04 points↑), and 24.64 points (2.19 points↑) in terms of mAP on C919, Roundabout, Trailer, and Boeing 777, respectively. The results of these experiments demonstrate that CBS is effective in improving the performance of the tail classes.

D. Comparisons With the SOTA

The comparison experiments of the PCLDet and other methods on Fair1M and OFSD are analyzed in this section. In comparison experiments, we compare the performance of the proposed PCLDet with well-known detection methods RetiaNet-O [22], Faster-RCNN-O [54], Gliding Vertex [16], S^2A-Net [18], RoI Transformer [14], Oriented RCNN [13], ReDet [17], and RB-FPN [55]. Among them, ReitaNet-O and Faster-RCNN-O are generally oriented object detection networks, and the rest approaches are designed for remote sensing object detection, where RB-FPN is designed for fine-grained remote sensing object detection.

1) Results on Fair1M: According to Table III, the PCLDet (ReDet) obtains superior performance with 43.5 points and is 1.5 points higher than the ReDet. The quantization results of RetiaNet-O and S^2A-Net are low, while the quantization results of Faster RCNN-O, Gliding Vertex, ReDet, and RoI Transformer are high. RetiaNet-O and S^2A-Net are one-stage detectors, and the framework of one-stage detectors limits their performance on the Fair1M dataset. Conversely, two-stage detectors (e.g., Faster RCNN-O, Gliding Vertex, and RoI Transformer) achieve better performance due to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

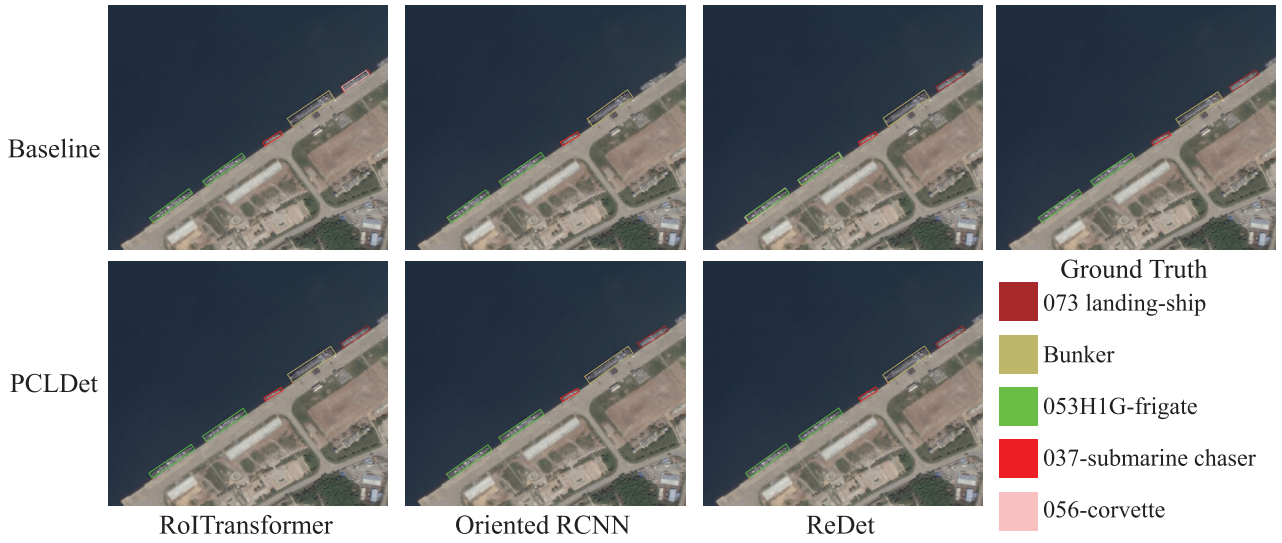


Fig. 6. Visualization of detection results on OFSD, using baseline (the first row) and the PCLDet (the second row).

inclusion of a region of interest generating stage. Based on high-precision two-stage detectors, the PCLDet achieves more discriminative feature extraction and the balanced sampling of categories. The comparison experiment results demonstrate the effectiveness of our method for fine-grained object detection in remote sensing images. In addition, to show the generality of our method, we select the SOTA two-stage oriented detectors ReDet, Oriented RCNN, and RoI Transformer as the baseline for comparison experiments. Our proposed PCLDet brings significant improvements to all listed two-stage oriented detectors. The PCLDet (RoI Transformer) and the PCLDet (Oriented RCNN) produce 1.02 points and 1.6 points improvement on RoI Transformer and Oriented RCNN, respectively.

2) Results on OFSD: The detection results on OFSD are shown in Table IV. As can be observed, our proposed PCLDet achieves the highest points in mAP on OFSD, which shows its effectiveness for fine-grained remote sensing object detection. We also designed a comparison experiment on different baselines. Experiment results demonstrate that our method can be plugged into any two-stage oriented detector and achieve remarkable improvements. Note that the PCLDet has a more significant improvement in OFSD than Fair1M. For example, the PCLDet (ReDet) improves by 2.17 points from ReDet on the OFSD and 1.5 points on the Fair1M. The reason that accounts for this difference is that Fair1M has multiple major classes (e.g., ship, car, and airplane), while OFSD has one major class. The OFSD dataset has higher interclass similarity compared to Fair1M. The PCLDet is designed to extract more discriminative features and works better in datasets with higher interclass similarity.

3) Qualitative Analysis: Fig. 4 shows the visual representation of the PCLDet on Fair1M. Since the ground truth of the Fair1M test set is not publicly available, we performed a qualitative analysis on the OFSD dataset. As shown in Fig. 5, the PCLDet can accurately detect some targets that cannot be correctly detected by compared methods, demonstrating the superiority of the PCLDet in the fine-grained remote sensing object detection task. For instance, in the first and third rows of Fig. 5, the PCLDet can accurately detect Nimitz Aircraft Carrier and Arleigh Burke-class Destroyer, which cannot be correctly detected by other methods. To demonstrate the generality of our approach, we also visualize the compared results of baseline and the PCLDet in Fig. 6. As displayed, the PCLDet is able to accurately detect targets that are misdetected or missed in the baseline methods: RoI Transformer, Oriented RCNN, and ReDet.

V. Conclusion

In this article, we propose the PCLDet for fine-grained object detection in remote sensing images. In the PCLDet, we propose the PCL module to extract the discriminative features and the CBS to alleviate the long-tail issue in remote sensing fine-grained object detection tasks. Extensive experiments show that the PCLDet yields a significant improvement in detection accuracy, demonstrating that our method achieves great performance in the task of fine-grained object detection in remote sensing images. Due to the frequency of object occurrence in real scenarios, the finite number of objects in particular fine-grained classes limits the detection performance. Hence, in our future work, we will turn our attention to few-shot fine-grained object detection in remote sensing images.

Acknowledgment

The authors would like to thank three anonymous reviewers and editors for their comments, which helped improve this article.

References

[1] L. Hou, K. Lu, and J. Xue, "Refined one-stage oriented object detection method for remote sensing images," *IEEE Trans. Image Process.*, vol. 31, pp. 1545–1558, 2022.

[2] L. Fang, Y. Jiang, Y. Yan, J. Yue, and Y. Deng, "Hyperspectral image instance segmentation using spectral–spatial feature pyramid network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502613.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

[3] Y. Liu, Q. Li, Y. Yuan, Q. Du, and Q. Wang, "ABNet: Adaptive balanced network for multiscale object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614914.

[4] W. Ma et al., "Feature split–merge–enhancement network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5616217.

[5] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[6] Q. Wang, J. Gao, and Y. Yuan, "A joint convolutional neural networks and context transfer for street scenes labeling," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 5, pp. 1457–1470, May 2018.

[7] B. Kalantar, S. B. Mansor, A. A. Halin, H. Z. M. Shafri, and M. Zand, "Multiple moving object detection from UAV videos using trajectories of matched regional adjacency graphs," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5198–5213, Sep. 2017.

[8] L. Wu, L. Fang, J. Yue, B. Zhang, P. Ghamisi, and M. He, "Deep bilateral filtering network for point-supervised semantic segmentation in remote sensing images," *IEEE Trans. Image Process.*, vol. 31, pp. 7419–7434, 2022.

[9] X. Han, Y. Zhong, and L. Zhang, "An efficient and robust integrated geospatial object detection framework for high spatial resolution remote sensing imagery," *Remote Sens.*, vol. 9, no. 7, p. 666, Jun. 2017.

[10] J. Yue, L. Fang, and M. He, "Spectral–spatial latent reconstruction for open-set hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5227–5241, 2022.

[11] T. Xu, X. Sun, W. Diao, L. Zhao, K. Fu, and H. Wang, "ASSD: Feature aligned single-shot detection for multiscale objects in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607117.

[12] D. Yu and S. Ji, "A new spatial-oriented object detection framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407416.

[13] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3500–3509.

[14] J. Ding, N. Xue, Y. Long, G. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2844–2853.

[15] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, Aug. 2021, vol. 35, no. 4, pp. 3163–3171.

[16] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.

[17] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2785–2794.

[18] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[19] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8231–8240.

[20] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[21] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, Oct. 2016, pp. 21–37.

[22] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[23] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[24] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6517–6525.

[25] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[26] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 6154–6162.

[27] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[28] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[29] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.

[30] G. Cheng et al., "Anchor-free oriented proposal generator for object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625411.

[31] L. Zeng et al., "Instance switching-based contrastive learning for fine-grained airplane detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Nov. 2022, Art. no. 5633416.

[32] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[33] W. Xiong, Z. Xiong, and Y. Cui, "An explainable attention network for fine-grained ship classification using remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620314.

[34] G. Sumbul, R. G. Cinbis, and S. Aksoy, "Multisource region attention network for fine-grained object recognition in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4929–4937, Jul. 2019.

[35] X. Zhang, Y. Lv, L. Yao, W. Xiong, and C. Fu, "A new benchmark and an attribute-guided multilevel feature representation network for fine-grained ship classification in optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1271–1285, Mar. 2020.

[36] L. Ouyang, L. Fang, and X. Ji, "Multigranularity self-attention network for fine-grained ship detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9722–9732, 2022.

[37] J. Ma et al., "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.

[38] S. Kong and C. Fowlkes, "Low-rank bilinear pooling for fine-grained classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7025–7034.

[39] S. Cai, W. Zuo, and L. Zhang, "Higher-order integration of hierarchical convolutional activations for fine-grained visual categorization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 511–520.

[40] C. Yu, X. Zhao, Q. Zheng, P. Zhang, and X. You, "Hierarchical bilinear pooling for fine-grained visual recognition," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, Sep. 2018, pp. 595–610.

[41] W. Ge, X. Lin, and Y. Yu, "Weakly supervised complementary parts models for fine-grained image classification from the bottom up," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3029–3038.

[42] Z. Wang, S. Wang, H. Li, Z. Dou, and J. Li, "Graph-propagation based correlation learning for weakly supervised fine-grained image classification," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12289–12296.

[43] N. Zhang, J. Donahue, R. Girshick, and T. Darrell, "Part-based R-CNNs for fine-grained category detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jun. 2014, pp. 834–849.

[44] H. Zheng, J. Fu, Z. Zha, and J. Luo, "Looking for the devil in the details: Learning trilinear attention sampling network for fine-grained image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5007–5016.

[45] J. Fu, H. Zheng, and T. Mei, "Look closer to see better: Recurrent attention convolutional neural network for fine-grained image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4476–4484.

[46] A. Behera, Z. Wharton, P. R. P. G. Hewage, and A. Bera, "Context-aware attentional pooling (cap) for fine-grained visual classification," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 2, pp. 929–937.

[47] M. Wu, C. Zhuang, M. Mossé, D. Yamins, and N. Goodman, "On mutual information in contrastive learning for visual representations," 2020, *arXiv:2005.13149*.

[48] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.

[49] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 815–823.
[50] J. Deng, J. Guo, J. Yang, A. Lattas, and S. Zafeiriou, "Variational prototype learning for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11901–11910.
[51] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7348–7358.
[52] P. Khosla et al., "Supervised contrastive learning," in *Proc. Neural Inf. Process. Syst.*, vol. 33, Jun. 2020, pp. 18661–18673.
[53] X. Li, J. Wu, Z. Sun, Z. Ma, J. Cao, and J. Xue, "BSNet: Bi-similarity network for few-shot fine-grained image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 1318–1331, 2021.
[54] G. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.
[55] J. Song, L. Miao, Q. Ming, Z. Zhou, and Y. Dong, "Fine-grained object detection in remote sensing images via adaptive label assignment and refined-balanced feature pyramid network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 71–82, 2023.



Lihan Ouyang received the B.S. degree from Shandong University, Weihai, China, in 2019. She is currently pursuing the Ph.D. degree with the Laboratory of Vision and Image Processing, Hunan University, Changsha, China.

Her research interests include remote sensing object detection and fine-grained object detection.



Guangmiao Guo received the B.S. degree from the Wuhan University of Technology University, Wuhan, China, in 2020. She is currently pursuing the master's degree with the Laboratory of Vision and Image Processing, Hunan University, Changsha, China.

Her research interests include remote sensing object detection.



Leyuan Fang (Senior Member, IEEE) received the Ph.D. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2015.

From September 2011 to September 2012, he was a Visiting Ph.D. Student with the Department of Ophthalmology, Duke University, Durham, NC, USA, supported by the China Scholarship Council. From August 2016 to September 2017, he was a Post-Doctoral Researcher with the Department of Biomedical Engineering, Duke University. He is currently a Professor with the College of Electrical and Information Engineering, Hunan University. His research interests include sparse representation and multiresolution analysis in remote sensing and medical image processing.

Dr. Fang was a recipient of the 2nd-Grade National Award at the Nature and Science Progress of China in 2019. He is an Associate Editor of IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Neurocomputing*.



Pedram Ghamisi (Senior Member, IEEE) received the B.Sc. degree in civil (survey) engineering from the Tehran South Campus of Azad University, Tehran, Iran, in 2008, the M.Sc. degree (Hons.) in remote sensing from the K. N. Toosi University of Technology, Tehran, in 2012, and the Ph.D. degree in electrical and computer engineering from the University of Iceland, University of Iceland, Reykjavík, Iceland, in 2015.

From 2013 to 2014, he was with the School of Geography, Planning and Environmental Management, The University of Queensland, Brisbane, QLD, Australia. He was a Post-Doctoral Research Fellow at the University of Iceland. He has been a Post-Doctoral Research Fellow at the Technical University of Munich, Munich, Germany, and Heidelberg University, Heidelberg, Germany, since 2015. He was a Research Scientist at the German Aerospace Center, Remote Sensing Technology Institute, Oberpfaffenhofen, Germany, from 2015 to 2018. He was the Head of the Machine Learning Group, Helmholtz-Zentrum Dresden-Rossendorf, Dresden, Germany. His research interests include interdisciplinary research on remote sensing and machine (deep) learning, image, and signal processing; and multisensory data fusion.

Dr. Ghamisi was a recipient of the prestigious "Alexander von Humboldt Fellowship" in 2015; the prestigious "High Potential Program" in 2018; the Best Researcher Award for M.Sc. students at the K. N. Toosi University of Technology from 2010 to 2011; the 2013 IEEE International Geoscience and Remote Sensing Symposium, Melbourne, in 2013; the IEEE Mikio Takagi Prize for winning the Student Paper Competition, competing with almost 70 submissions; and the Data Fusion Contest 2017 organized by the Image Analysis and Data Fusion Technical Committee of the Geoscience and Remote Sensing Society. He was the winner of the 2017 Best Reviewer Prize of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS. In 2016, he was selected as the Talented International Researcher by Iran's National Elites Foundation. He serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the *Remote Sensing*.



Jun Yue received the B.Eng. degree in geodesy from Wuhan University, Wuhan, China, in 2013, and the Ph.D. degree in geographic information system (GIS) from Peking University, Beijing, China, in 2018.

He is currently an Assistant Professor with the School of Automation, Central South University, Changsha, China. His research interests include satellite image understanding, pattern recognition, and few-shot learning.

Dr. Yue serves as a reviewer for IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING, *Information Fusion*, and *Information Sciences*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:38:33 UTC from IEEE Xplore. Restrictions apply.